Retrieval Architecture

Overview

The recall path is four retrievers racing, one fusion step, and a token budget. Everything runs in parallel, so the slowest leg determines the observed latency rather than the sum of all of them.

Reranking

A cross-encoder rescores the fused candidate set. It is the single largest quality improvement per engineering hour, and also the largest latency line, so it stays optional per request.

Consolidation

A nightly pass clusters near-duplicate memories and promotes recurring episodes into standing preferences, so the graph gets smaller and better rather than monotonically larger over time.

Conclusion

Atomicity is the argument for keeping vectors, the graph and the queue in one database rather than three services.